

Lagged Covariances

under Piecewise-Constant Means

Ishan Sengupta

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1 Setup and notation

Let $X_1, \dots, X_n \in \mathbb{R}^p$ with

$$X_t = \theta_t + \varepsilon_t,$$

where $\theta_t \in \mathbb{R}^p$ is a deterministic, piecewise-constant mean with finitely many change points, and the noise ε_t is i.i.d. across t with $\mathbb{E}[\varepsilon_t] = 0$ and $\text{Cov}(\varepsilon_t) = \Sigma$. We write $\varepsilon_{t,a}$, $\theta_{t,a}$, $X_{t,a}$ for the a -th coordinate at time t . Independence across data points gives us $\mathbb{E}[\varepsilon_{u,i}\varepsilon_{v,i}] = \sigma_i^2 \mathbf{1}\{u = v\}$. A *change point* is a gap between consecutive times at which the mean vector θ changes; at least one coordinate moves there, though a given coordinate need not.

Assumption 1 (Null model). $\Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_p^2)$; the coordinates of ε_t are mutually independent. Moreover, any three consecutive time points have at most one change point between them — equivalently, no two change points fall in adjacent gaps.

Remark. Assumption 1 gives us *cross-coordinate independence*, and thus $\mathbb{E}[\varepsilon_{u,i}\varepsilon_{v,j}] = \mathbb{E}[\varepsilon_{u,i}\varepsilon_{v,j}\varepsilon_{w,k}] = \mathbb{E}[\varepsilon_{u,i}\varepsilon_{v,j}\varepsilon_{w,k}\varepsilon_{x,l}] = 0$ whenever one coordinate index differs from all the others.

We call a difference of the noise at two time points, for a fixed coordinate, an *increment*: the increment in coordinate i over the time pair (α, α') is $\varepsilon_{\alpha,i} - \varepsilon_{\alpha',i}$; likewise $\theta_{\alpha,i} - \theta_{\alpha',i}$ is the mean jump in coordinate i over that time pair.

$$\begin{aligned} \widehat{s}_{t,(a,b)} &:= \frac{1}{2}(X_{t,a} - X_{t+1,a})(X_{t,b} - X_{t+1,b}) \\ &\quad + \frac{1}{2}(X_{t+1,a} - X_{t+2,a})(X_{t+1,b} - X_{t+2,b}) \\ &\quad - \frac{1}{2}(X_{t,a} - X_{t+2,a})(X_{t,b} - X_{t+2,b}). \end{aligned} \tag{1}$$

We call the three consecutive times $\{t, t+1, t+2\}$ on which $\widehat{s}_{t,(a,b)}$ is built its *window*.

2 The $Q + L + C$ decomposition

Insert $X_{u,i} = \theta_{u,i} + \varepsilon_{u,i}$ into (1) and collect terms by the number of noise factors they carry. This gives $\widehat{s}_{t,(a,b)} = Q_{t,(a,b)} + L_{t,(a,b)} + C_{t,(a,b)}$, with the quadratic part

$$\begin{aligned} Q_{t,(a,b)} &= \frac{1}{2}(\varepsilon_{t,a} - \varepsilon_{t+1,a})(\varepsilon_{t,b} - \varepsilon_{t+1,b}) \\ &\quad + \frac{1}{2}(\varepsilon_{t+1,a} - \varepsilon_{t+2,a})(\varepsilon_{t+1,b} - \varepsilon_{t+2,b}) \\ &\quad - \frac{1}{2}(\varepsilon_{t,a} - \varepsilon_{t+2,a})(\varepsilon_{t,b} - \varepsilon_{t+2,b}), \end{aligned} \tag{2}$$

the linear part

$$\begin{aligned} L_{t,(a,b)} &= \frac{1}{2}(\theta_{t,a} - \theta_{t+1,a})(\varepsilon_{t,b} - \varepsilon_{t+1,b}) + \frac{1}{2}(\varepsilon_{t,a} - \varepsilon_{t+1,a})(\theta_{t,b} - \theta_{t+1,b}) \\ &\quad + \frac{1}{2}(\theta_{t+1,a} - \theta_{t+2,a})(\varepsilon_{t+1,b} - \varepsilon_{t+2,b}) + \frac{1}{2}(\varepsilon_{t+1,a} - \varepsilon_{t+2,a})(\theta_{t+1,b} - \theta_{t+2,b}) \\ &\quad - \frac{1}{2}(\theta_{t,a} - \theta_{t+2,a})(\varepsilon_{t,b} - \varepsilon_{t+2,b}) - \frac{1}{2}(\varepsilon_{t,a} - \varepsilon_{t+2,a})(\theta_{t,b} - \theta_{t+2,b}), \end{aligned} \tag{3}$$

and the constant part

$$\begin{aligned} C_{t,(a,b)} &= \frac{1}{2}(\theta_{t,a} - \theta_{t+1,a})(\theta_{t,b} - \theta_{t+1,b}) \\ &\quad + \frac{1}{2}(\theta_{t+1,a} - \theta_{t+2,a})(\theta_{t+1,b} - \theta_{t+2,b}) \\ &\quad - \frac{1}{2}(\theta_{t,a} - \theta_{t+2,a})(\theta_{t,b} - \theta_{t+2,b}). \end{aligned} \tag{4}$$

Note. By Assumption 1 the window $\{t, t+1, t+2\}$ holds at most one change point, so $C_{t,(a,b)} = 0$; this is what makes $\hat{s}_{t,(a,b)}$ unbiased for $\sigma_{a,b}$.

Convention and symmetry. We label a pair by its two distinct coordinates in increasing order, so (a, b) always means $a < b$, and $\hat{s}_{t,(a,b)}$ is defined only for such pairs. The two coordinates enter (1) through the same expression — interchanging them leaves each product unchanged — so a pair is effectively an unordered set $\{a, b\}$, and every quantity built from $\hat{s}_{t,(a,b)}$ below depends only on that set, not on which coordinate is written first.

3 Reduction of the covariance at lag m

Fix a lag $m \geq 0$. Because $C_{t,(a,b)}$ and $C_{t+m,(c,d)}$ are constants,

$$\begin{aligned} \text{Cov}(\hat{s}_{t,(a,b)}, \hat{s}_{t+m,(c,d)}) &= \text{Cov}(Q_{t,(a,b)}, Q_{t+m,(c,d)}) + \text{Cov}(Q_{t,(a,b)}, L_{t+m,(c,d)}) \\ &\quad + \text{Cov}(L_{t,(a,b)}, Q_{t+m,(c,d)}) + \text{Cov}(L_{t,(a,b)}, L_{t+m,(c,d)}). \end{aligned}$$

Proposition 1 (The two cross terms vanish at every lag). For all $m \geq 0$, $\text{Cov}(Q_{t,(a,b)}, L_{t+m,(c,d)}) = \text{Cov}(L_{t,(a,b)}, Q_{t+m,(c,d)}) = 0$.

Proof. A term of $L_{t,(a,b)} Q_{t+m,(c,d)}$ is a deterministic mean jump times $\varepsilon_{p,i} \varepsilon_{q,c} \varepsilon_{w,d}$, with one noise factor from $L_{t,(a,b)}$ carrying a coordinate $i \in \{a, b\}$ and the two factors from $Q_{t+m,(c,d)}$ carrying the coordinates $c \neq d$. The three noise coordinates form the multiset $\{i, c, d\}$; since $c \neq d$, some coordinate occurs exactly once. By cross-coordinate independence the expectation factorizes and that lone factor contributes $\mathbb{E}[\varepsilon] = 0$. This argument counts coordinates, not times, so it is unaffected by the lag m . Hence $\mathbb{E}[L_{t,(a,b)} Q_{t+m,(c,d)}] = 0$, and since $\mathbb{E}[L_{t,(a,b)}] = 0$ the covariance vanishes; the other term is identical by symmetry. $\square$

$$\text{Cov}(\hat{s}_{t,(a,b)}, \hat{s}_{t+m,(c,d)}) = \underbrace{\text{Cov}(Q_{t,(a,b)}, Q_{t+m,(c,d)})}_{\text{quadratic part}} + \underbrace{\text{Cov}(L_{t,(a,b)}, L_{t+m,(c,d)})}_{\text{linear part}} \quad (m \geq 0).$$

4 The quadratic part $\text{Cov}(Q_{t,(a,b)}, Q_{t+m,(c,d)})$

Fix a pair of coordinates $a < b$. Here $\sigma_{a,b}$ is the (a, b) entry of the covariance matrix Σ ; it lies off the diagonal of Σ , and under the null Σ is diagonal, so $\sigma_{a,b} = 0$, and likewise $\sigma_{c,d} = 0$ for the second pair $c < d$. Hence $\mathbb{E}[Q_{t,(a,b)}] = \mathbb{E}[Q_{t+m,(c,d)}] = 0$ and

$$\text{Cov}(Q_{t,(a,b)}, Q_{t+m,(c,d)}) = \mathbb{E}[Q_{t,(a,b)} Q_{t+m,(c,d)}].$$

$Q_{t,(a,b)}$ contains three terms. Each term is a product of two increments over the same time pair — one in coordinate a , one in coordinate b — so each term is labelled by that single time pair. The three time pairs of $Q_{t,(a,b)}$ are $(t, t+1)$, $(t+1, t+2)$, $(t, t+2)$; write s_I for the sign of the term with time pair I , so $s_{(t,t+1)} = s_{(t+1,t+2)} = +1$ and $s_{(t,t+2)} = -1$ (the signs in (2)). By the same reading the three time pairs of $Q_{t+m,(c,d)}$ are $(t+m, t+m+1)$, $(t+m+1, t+m+2)$, $(t+m, t+m+2)$, with the same signs s_J . Thus $Q_{t,(a,b)}$ uses the times $t, t+1, t+2$ and $Q_{t+m,(c,d)}$

uses $t+m, t+m+1, t+m+2$: the two windows coincide when $m=0$, share the times $t+1, t+2$ when $m=1$, and only the time $t+2$ when $m=2$.

The product $Q_{t,(a,b)} Q_{t+m,(c,d)}$ has $3 \times 3 = 9$ terms. We take one such term: a term of $Q_{t,(a,b)}$ whose time pair is $I = (\alpha, \alpha')$, times a term of $Q_{t+m,(c,d)}$ whose time pair is $J = (\gamma, \gamma')$. Summing over the nine choices,

$$\mathbb{E}[Q_{t,(a,b)} Q_{t+m,(c,d)}] = \frac{1}{4} \sum_{I,J} s_I s_J \mathbb{E}[(\varepsilon_{\alpha,a} - \varepsilon_{\alpha',a})(\varepsilon_{\alpha,b} - \varepsilon_{\alpha',b})(\varepsilon_{\gamma,c} - \varepsilon_{\gamma',c})(\varepsilon_{\gamma,d} - \varepsilon_{\gamma',d})].$$

■ When the two coordinate pairs differ.

Take one term. Its expectation is over four noise factors, in coordinates a, b (from the $Q_{t,(a,b)}$ side) and c, d (from the $Q_{t+m,(c,d)}$ side). Variables with different coordinates are mutually independent, so the expectation splits into a product over the distinct coordinates present, and any coordinate appearing a single time contributes a factor $\mathbb{E}[\varepsilon] = 0$. Since $a \neq b$ and $c \neq d$, every coordinate appears at least twice only when $(a, b) = (c, d)$; the expectation is therefore zero unless the two coordinate pairs coincide.

■ When the two coordinate pairs coincide.

Set $(a, b) = (c, d)$. The four factors are then two increments in coordinate a (over I and over J) and two in coordinate b (over I and over J). As a and b are independent, the expectation factorizes as

$$\mathbb{E}[(\varepsilon_{\alpha,a} - \varepsilon_{\alpha',a})(\varepsilon_{\gamma,a} - \varepsilon_{\gamma',a})] \cdot \mathbb{E}[(\varepsilon_{\alpha,b} - \varepsilon_{\alpha',b})(\varepsilon_{\gamma,b} - \varepsilon_{\gamma',b})].$$

Each factor is the covariance of the increment over I and the increment over J for the same coordinate. Using $\mathbb{E}[\varepsilon_{p,i} \varepsilon_{q,i}] = \sigma_i^2 \mathbf{1}\{p = q\}$,

$$\mathbb{E}[(\varepsilon_{\alpha,i} - \varepsilon_{\alpha',i})(\varepsilon_{\gamma,i} - \varepsilon_{\gamma',i})] = \sigma_i^2 g_{IJ}, \quad g_{IJ} := \mathbf{1}\{\alpha = \gamma\} - \mathbf{1}\{\alpha = \gamma'\} - \mathbf{1}\{\alpha' = \gamma\} + \mathbf{1}\{\alpha' = \gamma'\}.$$

In particular $g_{II} = 2$: a single increment has variance $2\sigma_i^2$. Both coordinates run over the same time pairs I and J , so each contributes the same g_{IJ} , and the term equals $\sigma_a^2 \sigma_b^2 g_{IJ}^2$. Hence

$$\mathbb{E}[Q_{t,(a,b)} Q_{t+m,(c,d)}] = \frac{1}{4} \sigma_a^2 \sigma_b^2 \sum_{I,J} s_I s_J g_{IJ}^2.$$

■ Lag $m = 0$.

When $m = 0$ the two statistics sit on the same window, so I and J run over the same three time pairs $(t, t+1), (t+1, t+2), (t, t+2)$ and the increments can coincide ($g_{II} = 2$). The g_{IJ} are

g_{IJ}	$(t, t+1)$	$(t+1, t+2)$	$(t, t+2)$
$(t, t+1)$	2	-1	1
$(t+1, t+2)$	-1	2	1
$(t, t+2)$	1	1	2

Squaring and attaching the signs, the row sums are

$$\underbrace{(+4) + (+1) + (-1)}_{I=(t, t+1)} + \underbrace{(+1) + (+4) + (-1)}_{I=(t+1, t+2)} + \underbrace{(-1) + (-1) + (+4)}_{I=(t, t+2)} = 4 + 4 + 2 = 10.$$

Hence $\mathbb{E}[Q_{t,(a,b)} Q_{t,(c,d)}] = \frac{1}{4} \cdot 10 \sigma_a^2 \sigma_b^2 = \frac{5}{2} \sigma_a^2 \sigma_b^2$, the contemporaneous variance.

■ **Lag $m = 1$.**

The time pairs of $Q_{t+1,(c,d)}$ are $(t+1, t+2)$, $(t+2, t+3)$, $(t+1, t+3)$, and the g_{IJ} are

g_{IJ}	$(t+1, t+2)$	$(t+2, t+3)$	$(t+1, t+3)$
$(t, t+1)$	-1	0	-1
$(t+1, t+2)$	2	-1	1
$(t, t+2)$	1	-1	0

Squaring each entry and attaching the signs $s_I s_J$ (both sign patterns are $+, +, -$), the row sums are

$$\underbrace{(+1) + (+0) + (-1)}_{I=(t, t+1)} + \underbrace{(+4) + (+1) + (-1)}_{I=(t+1, t+2)} + \underbrace{(-1) + (-1) + (+0)}_{I=(t, t+2)} = 0 + 4 - 2 = 2.$$

The single large contribution is $I = J = (t+1, t+2)$, the increment over the time pair shared by the two windows, where $g_{IJ} = 2$. Hence $\mathbb{E}[Q_{t,(a,b)} Q_{t+1,(c,d)}] = \frac{1}{4} \cdot 2 \sigma_a^2 \sigma_b^2 = \frac{1}{2} \sigma_a^2 \sigma_b^2$.

■ **Lag $m = 2$.**

Now $Q_{t+2,(c,d)}$ has time pairs $(t+2, t+3)$, $(t+3, t+4)$, $(t+2, t+4)$, and the two windows share only the single time $t+2$, so no time pair of $Q_{t,(a,b)}$ can equal a time pair of $Q_{t+2,(c,d)}$. The g_{IJ} are

g_{IJ}	$(t+2, t+3)$	$(t+3, t+4)$	$(t+2, t+4)$
$(t, t+1)$	0	0	0
$(t+1, t+2)$	-1	0	-1
$(t, t+2)$	-1	0	-1

Squaring and attaching the signs, the row sums are

$$\underbrace{(+0) + (+0) + (-0)}_{I=(t, t+1)} + \underbrace{(+1) + (+0) + (-1)}_{I=(t+1, t+2)} + \underbrace{(-1) + (-0) + (+1)}_{I=(t, t+2)} = 0.$$

Hence $\mathbb{E}[Q_{t,(a,b)} Q_{t+2,(c,d)}] = 0$.

$$\text{Cov}(Q_{t,(a,b)}, Q_{t+m,(c,d)}) = \begin{cases} \frac{5}{2} \sigma_a^2 \sigma_b^2, & m = 0 \text{ and } (c, d) = (a, b), \\ \frac{1}{2} \sigma_a^2 \sigma_b^2, & m = 1 \text{ and } (c, d) = (a, b), \\ 0, & \text{otherwise.} \end{cases}$$

The quadratic part is **one-dependent**: nonzero only at lags 0 and 1, and only when the two coordinate pairs coincide.

5 The linear part $\text{Cov}(L_{t,(a,b)}, L_{t+m,(c,d)})$

Since $\mathbb{E}[L_{t,(a,b)}] = 0$ and $\mathbb{E}[L_{t+m,(c,d)}] = 0$, the linear part equals the product moment

$$\text{Cov}(L_{t,(a,b)}, L_{t+m,(c,d)}) = \mathbb{E}[L_{t,(a,b)} L_{t+m,(c,d)}].$$

We expand the product $L_{t,(a,b)} L_{t+m,(c,d)}$ into single terms and find which terms have nonzero expectation.

A single term of the product. By (3), each of the six terms of $L_{t,(a,b)}$ is a mean jump in one of the two coordinates a, b times a noise increment in the other, both over one time pair; the six terms of $L_{t+m,(c,d)}$ have the same form in the coordinates c, d . Fix one term of $L_{t,(a,b)}$, over the time pair $I = (\alpha, \alpha')$, and one term of $L_{t+m,(c,d)}$, over the time pair $J = (\gamma, \gamma')$. To be definite, take the term of $L_{t,(a,b)}$ carrying the mean jump in coordinate a (and hence the noise increment in coordinate b) and the term of $L_{t+m,(c,d)}$ carrying the mean jump in coordinate c (and hence the increment in coordinate d); their product is

$$\frac{1}{4} s_I s_J \underbrace{(\theta_{\alpha,a} - \theta_{\alpha',a})(\theta_{\gamma,c} - \theta_{\gamma',c})}_{\text{deterministic mean jumps}} \underbrace{(\varepsilon_{\alpha,b} - \varepsilon_{\alpha',b})(\varepsilon_{\gamma,d} - \varepsilon_{\gamma',d})}_{\text{noise increments}},$$

where $s_I, s_J \in \{+, +, -\}$ are the signs attached to the time pairs I and J . Its expectation is the deterministic factor times $\mathbb{E}[(\varepsilon_{\alpha,b} - \varepsilon_{\alpha',b})(\varepsilon_{\gamma,d} - \varepsilon_{\gamma',d})]$, the covariance of the increment in coordinate b over the time pair I with the increment in coordinate d over the time pair J . The remaining three terms of the product carry the two increments in the other combinations of one coordinate from $\{a, b\}$ and one coordinate from $\{c, d\}$, and have the same form.

Which terms have nonzero expectation. In each such term the two increments lie in one coordinate from $\{a, b\}$, call it i , and one coordinate from $\{c, d\}$, call it i' . By cross-coordinate independence,

$$\mathbb{E}[(\varepsilon_{\alpha,i} - \varepsilon_{\alpha',i})(\varepsilon_{\gamma,i'} - \varepsilon_{\gamma',i'})] = 0 \quad \text{whenever } i \neq i',$$

so a term has nonzero expectation only when $i = i'$ — that is, only when its two increments lie in one and the same coordinate. Writing that common coordinate as i ,

$$\mathbb{E}[(\varepsilon_{\alpha,i} - \varepsilon_{\alpha',i})(\varepsilon_{\gamma,i} - \varepsilon_{\gamma',i})] = \sigma_i^2 g_{IJ},$$

with the same coefficient $g_{IJ} = \mathbf{1}\{\alpha = \gamma\} - \mathbf{1}\{\alpha = \gamma'\} - \mathbf{1}\{\alpha' = \gamma\} + \mathbf{1}\{\alpha' = \gamma'\}$ as for the quadratic part. Such a common coordinate exists only if $\{a, b\}$ and $\{c, d\}$ share a coordinate, which gives the three cases below.

■ (i) **Disjoint pairs.**

If $\{a, b\} \cap \{c, d\} = \emptyset$, then in every term the increment from $L_{t,(a,b)}$ (in coordinate a or b) and the increment from $L_{t+m,(c,d)}$ (in coordinate c or d) lie in different coordinates. By the display above every term has expectation 0, so $\text{Cov}(L_{t,(a,b)}, L_{t+m,(c,d)}) = 0$.

■ (ii) **One shared coordinate.**

If $\{a, b\} \cap \{c, d\} = \{j\}$, write the two partners as r and s , where $\{a, b\} = \{j, r\}$ and $\{c, d\} = \{j, s\}$. Coordinate j is the only coordinate common to both pairs, so a term has nonzero expectation only when both its increments lie in coordinate j ; by the description of a single term, this happens exactly when the term of $L_{t,(a,b)}$ carries the mean jump in the partner r (increment in j) and the term of $L_{t+m,(c,d)}$ carries the mean jump in the partner s (increment in j). Summing these terms over the three time pairs I of the first window and the three time pairs J of the second,

$$\mathbb{E}[L_{t,(a,b)} L_{t+m,(c,d)}] = \frac{1}{4} \sigma_j^2 \sum_{I,J} s_I s_J g_{IJ} (\theta_{\alpha,r} - \theta_{\alpha',r})(\theta_{\gamma,s} - \theta_{\gamma',s}), \quad (5)$$

where $I = (\alpha, \alpha')$, $J = (\gamma, \gamma')$, and $\theta_{\alpha,r} - \theta_{\alpha',r}$ is the mean jump in the partner r over the time pair I .

■ (iii) Equal pairs, $(a, b) = (c, d)$.

Now $\{a, b\}$ and $\{c, d\}$ share both coordinates, so a term has nonzero expectation when both its increments lie in coordinate a , or both lie in coordinate b . Summing the terms with both increments in coordinate a (mean jump in the partner b) and the terms with both increments in coordinate b (mean jump in the partner a),

$$\mathbb{E}[L_{t,(a,b)} L_{t+m,(a,b)}] = \frac{1}{4} \sum_{I,J} s_I s_J g_{IJ} \left[\sigma_a^2 (\theta_{\alpha,b} - \theta_{\alpha',b}) (\theta_{\gamma,b} - \theta_{\gamma',b}) + \sigma_b^2 (\theta_{\alpha,a} - \theta_{\alpha',a}) (\theta_{\gamma,a} - \theta_{\gamma',a}) \right]. \quad (6)$$

Evaluating the sum (5). We evaluate (5) for one shared coordinate. Let j be the coordinate shared by the two pairs (a, b) and (c, d) ; write r for the other coordinate of the first pair and s for the other coordinate of the second. The pairs stay in increasing order, $a < b$ and $c < d$, so j , r , and s may occupy either slot, but the computation refers only to which coordinate carries the noise increment, never to slot order. By case (ii), a term of $L_{t,(a,b)}$ contributes to this covariance only when its noise increment is in the shared coordinate j (equivalently, its mean jump is in the partner r), and a term of $L_{t+m,(c,d)}$ contributes only when its increment is in j (its jump in the partner s); the terms of $L_{t,(a,b)}$ whose increment is in r , and of $L_{t+m,(c,d)}$ whose increment is in s , contribute nothing.

Two facts then settle every placement of the change points. *Reduction:* by Assumption 1 a window holds at most one change point. A window $\{u, u+1, u+2\}$ has two gaps, $(u, u+1)$ and $(u+1, u+2)$; suppose the statistic's change point lies in one of them, say (α, α') . Then the statistic's three contributing terms collapse to a single noise increment over the remaining gap — the one without the change point — carrying the coefficient $\frac{1}{2}(\theta_{\alpha,x} - \theta_{\alpha',x})$, where x is the partner. If the partner does not move at that change point, this coefficient is 0 and the contributing part vanishes. *Overlap:* the covariance of the two reduced increments equals $\sigma_j^2 g_{IJ}$, the coefficient g_{IJ} from the quadratic part now taken for those two increments (with I, J their time pairs): $g_{IJ} = 2$ if they are the same increment, $g_{IJ} = -1$ if they are adjacent (sharing one time, which enters the two increments with opposite signs), and $g_{IJ} = 0$ if they are disjoint. Finally, Assumption 1 forbids the two change points from lying in adjacent gaps, as that would place two change points in a single window.

■ Lag 0.

Here $L_{t,(a,b)}$ and $L_{t,(c,d)}$ are both on the window $\{t, t+1, t+2\}$, which holds at most one change point; suppose it lies in $(t, t+1)$. The three contributing terms of $L_{t,(a,b)}$ carry the mean jump in r over the time pairs $(t, t+1)$, $(t+1, t+2)$, $(t, t+2)$ with signs $+, +, -$. Since the window's only change point is in $(t, t+1)$, the mean of r is constant across $(t+1, t+2)$, so its jump there is $\theta_{t+1,r} - \theta_{t+2,r} = 0$ and the middle term drops; its jump over $(t, t+2)$ equals its jump over $(t, t+1)$, $\theta_{t,r} - \theta_{t+2,r} = \theta_{t,r} - \theta_{t+1,r}$. The two surviving terms share the factor $\frac{1}{2}(\theta_{t,r} - \theta_{t+1,r})$, and their noise parts cancel to a single increment over $(t+1, t+2)$:

$$\frac{1}{2}(\theta_{t,r} - \theta_{t+1,r})(\varepsilon_{t,j} - \varepsilon_{t+1,j}) - \frac{1}{2}(\theta_{t,r} - \theta_{t+1,r})(\varepsilon_{t,j} - \varepsilon_{t+2,j}) = \frac{1}{2}(\theta_{t,r} - \theta_{t+1,r})(\varepsilon_{t+2,j} - \varepsilon_{t+1,j}).$$

The same reduction of $L_{t,(c,d)}$ gives $\frac{1}{2}(\theta_{t,s} - \theta_{t+1,s})(\varepsilon_{t+2,j} - \varepsilon_{t+1,j})$, the same increment ($g_{IJ} = 2$). Multiplying and taking the expectation,

$$\begin{aligned} \mathbb{E}[L_{t,(a,b)} L_{t,(c,d)}] &= \frac{1}{4}(\theta_{t,r} - \theta_{t+1,r})(\theta_{t,s} - \theta_{t+1,s}) \mathbb{E}[(\varepsilon_{t+2,j} - \varepsilon_{t+1,j})^2] \\ &= \frac{1}{2} \sigma_j^2 (\theta_{t,r} - \theta_{t+2,r})(\theta_{t,s} - \theta_{t+2,s}), \end{aligned}$$

using $\mathbb{E}[(\varepsilon_{t+2,j} - \varepsilon_{t+1,j})^2] = 2\sigma_j^2$.

Other placements. The window's single change point may lie in either gap, $(t, t+1)$ or $(t+1, t+2)$, with the same value, since the full-window jump $\theta_{t,\cdot} - \theta_{t+2,\cdot}$ records it wherever it lies; if the window has no change point, the linear part is 0.

■ Lag 1.

Now $L_{t,(a,b)}$ is on $\{t, t+1, t+2\}$ and $L_{t+1,(c,d)}$ on $\{t+1, t+2, t+3\}$. Suppose there are change points in $(t, t+1)$ and $(t+2, t+3)$; these gaps are two apart, so Assumption 1 permits them — the window $\{t, t+1, t+2\}$ contains only the one in $(t, t+1)$, and the window $\{t+1, t+2, t+3\}$ only the one in $(t+2, t+3)$. Each change point moves the whole vector; of these movements the covariance reads only r 's jump across $(t, t+1)$ and s 's jump across $(t+2, t+3)$. By the reduction, $L_{t,(a,b)}$ gives $\frac{1}{2}(\theta_{t,r} - \theta_{t+1,r})(\varepsilon_{t+2,j} - \varepsilon_{t+1,j})$ over $(t+1, t+2)$, and $L_{t+1,(c,d)}$ gives $\frac{1}{2}(\theta_{t+2,s} - \theta_{t+3,s})(\varepsilon_{t+2,j} - \varepsilon_{t+1,j})$ over the same gap $(t+1, t+2)$ ($g_{IJ} = 2$). Hence

$$\begin{aligned}\mathbb{E}[L_{t,(a,b)} L_{t+1,(c,d)}] &= \frac{1}{4}(\theta_{t,r} - \theta_{t+1,r})(\theta_{t+2,s} - \theta_{t+3,s}) \mathbb{E}[(\varepsilon_{t+2,j} - \varepsilon_{t+1,j})^2] \\ &= \frac{1}{2} \sigma_j^2 (\theta_{t,r} - \theta_{t+1,r})(\theta_{t+2,s} - \theta_{t+3,s}).\end{aligned}$$

Other placements. A change point in $(t, t+1)$ with one in $(t+1, t+2)$, or one in $(t+1, t+2)$ with one in $(t+2, t+3)$, is excluded: the two gaps are adjacent, so both change points would fall in one window. A single change point in the shared gap $(t+1, t+2)$ reduces $L_{t,(a,b)}$ and $L_{t+1,(c,d)}$ to increments over $(t, t+1)$ and $(t+2, t+3)$, which are disjoint ($g_{IJ} = 0$), so the covariance is 0. Thus only a change point in $(t, t+1)$ paired with one in $(t+2, t+3)$ contributes.

■ Lag 2.

Now $L_{t,(a,b)}$ is on $\{t, t+1, t+2\}$ and $L_{t+2,(c,d)}$ on $\{t+2, t+3, t+4\}$, and the two windows meet only at the single time $t+2$. Suppose there are change points in $(t, t+1)$ and $(t+3, t+4)$. Each moves the whole vector; of these movements the covariance reads only r 's jump across $(t, t+1)$ and s 's jump across $(t+3, t+4)$. By the reduction, $L_{t,(a,b)}$ gives $\frac{1}{2}(\theta_{t,r} - \theta_{t+1,r})(\varepsilon_{t+2,j} - \varepsilon_{t+1,j})$ over $(t+1, t+2)$, and $L_{t+2,(c,d)}$ gives $\frac{1}{2}(\theta_{t+3,s} - \theta_{t+4,s})(\varepsilon_{t+3,j} - \varepsilon_{t+2,j})$ over $(t+2, t+3)$. These two increments are adjacent — they share the single time $t+2$, which enters the first with a plus sign and the second with a minus sign, so $g_{IJ} = -1$. Hence

$$\begin{aligned}\mathbb{E}[L_{t,(a,b)} L_{t+2,(c,d)}] &= \frac{1}{4}(\theta_{t,r} - \theta_{t+1,r})(\theta_{t+3,s} - \theta_{t+4,s}) \mathbb{E}[(\varepsilon_{t+2,j} - \varepsilon_{t+1,j})(\varepsilon_{t+3,j} - \varepsilon_{t+2,j})] \\ &= -\frac{1}{4} \sigma_j^2 (\theta_{t,r} - \theta_{t+1,r})(\theta_{t+3,s} - \theta_{t+4,s}).\end{aligned}$$

Other placements. Every other placement gives 0 or is excluded. A change point in $(t, t+1)$ with one in $(t+2, t+3)$: the reduced increments lie over $(t+1, t+2)$ and $(t+3, t+4)$, which are disjoint ($g_{IJ} = 0$), so the covariance is 0 — this is the placement with a change point in the second window's leading gap $(t+2, t+3)$. A change point in $(t+1, t+2)$ with one in $(t+3, t+4)$: the reduced increments lie over $(t, t+1)$ and $(t+2, t+3)$, again disjoint, so 0. A change point in $(t+1, t+2)$ with one in $(t+2, t+3)$ is excluded (adjacent gaps). Thus at lag 2 only a change point in $(t, t+1)$ paired with one in $(t+3, t+4)$ contributes.

■ Equal pairs.

When $(a, b) = (c, d)$, a surviving term of the product $L_{t,(a,b)} L_{t+m,(a,b)}$ has both its noise increments in coordinate a or both in coordinate b (case (iii)); a term with one increment in coordinate a and the other in coordinate b has its two increments in different coordinates and, by cross-coordinate independence, expectation 0. Each surviving family is exactly the one-shared computation above with the shared coordinate and partners specialized: the “both in a ” family takes $j = a$ and $r = s = b$, and the “both in b ” family takes $j = b$ and $r = s = a$. The change-point placements

and reductions of the lag-0, lag-1, lag-2 cases therefore carry over unchanged — in the “both in a ” family coordinate b plays the partner role in *both* statistics, so at lag 1 a change point in $(t, t+1)$ and one in $(t+2, t+3)$ each move b , permitted as those gaps are two apart. Adding the two families gives the equal-pairs entries in the box.

One shared coordinate j , with partners r and s :

$$\text{Cov}(L_{t,(a,b)}, L_{t,(c,d)}) = \frac{1}{2} \sigma_j^2 (\theta_{t,r} - \theta_{t+2,r})(\theta_{t,s} - \theta_{t+2,s}),$$

$$\text{Cov}(L_{t,(a,b)}, L_{t+1,(c,d)}) = \frac{1}{2} \sigma_j^2 (\theta_{t,r} - \theta_{t+1,r})(\theta_{t+2,s} - \theta_{t+3,s}),$$

$$\text{Cov}(L_{t,(a,b)}, L_{t+2,(c,d)}) = -\frac{1}{4} \sigma_j^2 (\theta_{t,r} - \theta_{t+1,r})(\theta_{t+3,s} - \theta_{t+4,s}).$$

Equal pairs, $(a, b) = (c, d)$:

$$\text{Cov}(L_{t,(a,b)}, L_{t,(a,b)}) = \frac{1}{2} \sigma_a^2 (\theta_{t,b} - \theta_{t+2,b})^2 + \frac{1}{2} \sigma_b^2 (\theta_{t,a} - \theta_{t+2,a})^2,$$

$$\begin{aligned} \text{Cov}(L_{t,(a,b)}, L_{t+1,(a,b)}) &= \frac{1}{2} \sigma_a^2 (\theta_{t,b} - \theta_{t+1,b})(\theta_{t+2,b} - \theta_{t+3,b}) \\ &\quad + \frac{1}{2} \sigma_b^2 (\theta_{t,a} - \theta_{t+1,a})(\theta_{t+2,a} - \theta_{t+3,a}), \end{aligned}$$

$$\begin{aligned} \text{Cov}(L_{t,(a,b)}, L_{t+2,(a,b)}) &= -\frac{1}{4} \sigma_a^2 (\theta_{t,b} - \theta_{t+1,b})(\theta_{t+3,b} - \theta_{t+4,b}) \\ &\quad - \frac{1}{4} \sigma_b^2 (\theta_{t,a} - \theta_{t+1,a})(\theta_{t+3,a} - \theta_{t+4,a}). \end{aligned}$$

Disjoint pairs: 0 at every lag.

6 The assembled lagged autocovariance

Collecting the two parts at each lag gives the following.

Theorem 1 (Lag-0, lag-1 and lag-2 autocovariance). Assume the null model (Assumption 1), and take $a < b, c < d$.

Lag 0. When $(c, d) = (a, b)$,

$$\begin{aligned} \text{Cov}(\hat{s}_{t,(a,b)}, \hat{s}_{t,(a,b)}) &= \frac{5}{2} \sigma_a^2 \sigma_b^2 + \frac{1}{2} \sigma_a^2 (\theta_{t,b} - \theta_{t+2,b})^2 \\ &\quad + \frac{1}{2} \sigma_b^2 (\theta_{t,a} - \theta_{t+2,a})^2; \end{aligned}$$

when $\{a, b\} \cap \{c, d\} = \{j\}$ with partners r, s ,

$$\text{Cov}(\hat{s}_{t,(a,b)}, \hat{s}_{t,(c,d)}) = \frac{1}{2} \sigma_j^2 (\theta_{t,r} - \theta_{t+2,r})(\theta_{t,s} - \theta_{t+2,s});$$

and the covariance is 0 when the pairs are disjoint.

Lag 1. When $(c, d) = (a, b)$,

$$\begin{aligned} \text{Cov}(\hat{s}_{t,(a,b)}, \hat{s}_{t+1,(a,b)}) &= \frac{1}{2} \sigma_a^2 \sigma_b^2 + \frac{1}{2} \sigma_a^2 (\theta_{t,b} - \theta_{t+1,b})(\theta_{t+2,b} - \theta_{t+3,b}) \\ &\quad + \frac{1}{2} \sigma_b^2 (\theta_{t,a} - \theta_{t+1,a})(\theta_{t+2,a} - \theta_{t+3,a}); \end{aligned}$$

when $\{a, b\} \cap \{c, d\} = \{j\}$ with partners r, s ,

$$\text{Cov}(\hat{s}_{t,(a,b)}, \hat{s}_{t+1,(c,d)}) = \frac{1}{2} \sigma_j^2 (\theta_{t,r} - \theta_{t+1,r})(\theta_{t+2,s} - \theta_{t+3,s});$$

and the covariance is 0 when the pairs are disjoint.

Lag 2 (no quadratic part). When $(c, d) = (a, b)$,

$$\begin{aligned} \text{Cov}(\widehat{s}_{t,(a,b)}, \widehat{s}_{t+2,(a,b)}) &= -\frac{1}{4} \sigma_a^2 (\theta_{t,b} - \theta_{t+1,b})(\theta_{t+3,b} - \theta_{t+4,b}) \\ &\quad - \frac{1}{4} \sigma_b^2 (\theta_{t,a} - \theta_{t+1,a})(\theta_{t+3,a} - \theta_{t+4,a}); \end{aligned}$$

when $\{a, b\} \cap \{c, d\} = \{j\}$ with partners r, s ,

$$\text{Cov}(\widehat{s}_{t,(a,b)}, \widehat{s}_{t+2,(c,d)}) = -\frac{1}{4} \sigma_j^2 (\theta_{t,r} - \theta_{t+1,r})(\theta_{t+3,s} - \theta_{t+4,s});$$

and 0 when the pairs are disjoint.

For equal pairs $(c, d) = (a, b)$, the lag-0 covariance is the contemporaneous variance — the quadratic part $\frac{5}{2} \sigma_a^2 \sigma_b^2$ plus the linear-part squares; at lag 1 the quadratic part drops to $\frac{1}{2} \sigma_a^2 \sigma_b^2$; and at lag 2 it is gone (one-dependent), leaving only the linear term. For a single shared coordinate only the linear term is present, at all three lags, and disjoint pairs give 0 throughout.